

Medication Administration & Near-Miss Safety

Synthetic Data Case Study

Synthetic Australian Hospital Operations Dataset Bundle

Executive Summary

This case study walks through a sample analysis of the Medication Administration & Near-Miss Safety dataset — 32,000 synthetic medication administration events across 8 fictional facilities and 10 ward/unit types. It demonstrates the kind of shift-pattern, staffing-context, and risk-profiling analysis this dataset supports, using the included dashboard and notebook. All figures below are calculated directly from the dataset.

⚠ This is a synthetic dataset. Every figure below describes a simulated pattern, not a real-world clinical finding.

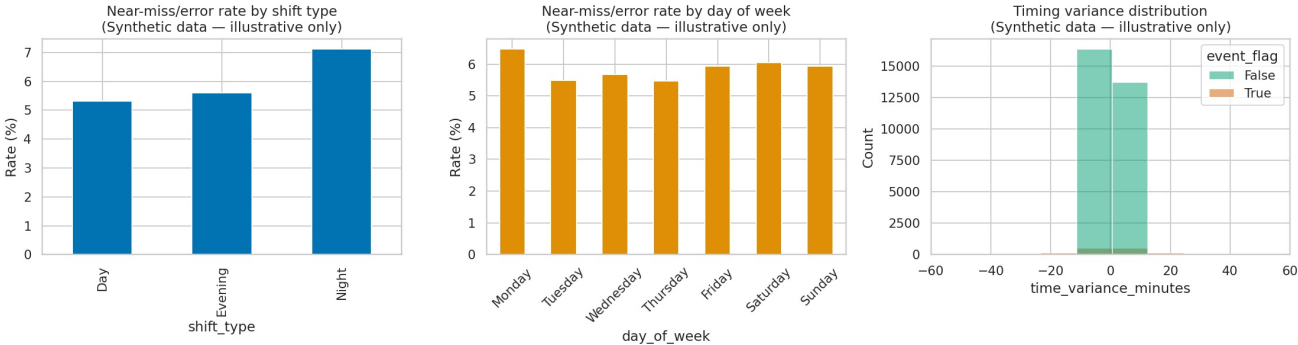
Dataset Snapshot

- 32,000 medication administration events, 22 columns
- 8 fictional facilities, 10 ward/unit types
- Modelled near-miss/error rate: 5.87% overall
- Date range: 2025-01-01 to 2025-12-30

Key Findings (in this synthetic model)

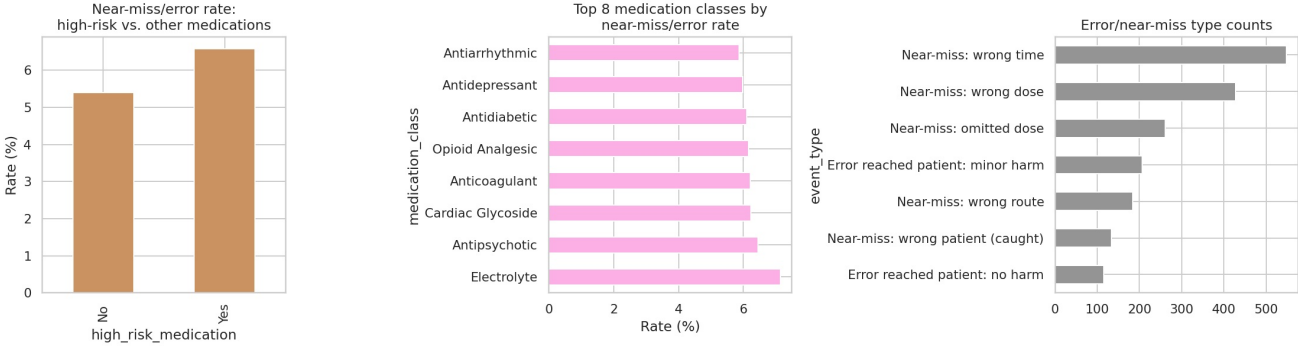
- Night shift carries the highest modelled incident rate — 7.12%, compared with 5.31% on day shift and 5.61% on evening shift.
- High-risk medications show a modestly higher modelled incident rate — 6.58%, versus 5.40% for other medications.
- The Surgical Ward shows the highest modelled incident rate among fictional units at approximately 6.36% — a designed pattern, not a real-world safety signal.
- Staff experience and staffing ratio both move the modelled incident rate in the expected direction: less experienced staff and higher patient-to-nurse ratios correlate with more modelled incidents, by design.

Visual Highlights



Time and staffing patterns

Modelled near-miss/error rate by shift, day of week, and timing variance — night shift stands out as the highest-rate category.



Medication risk context

High-risk vs. other medications, top medication classes by rate, and error/near-miss type counts.

Methodology Note

Near-miss/error likelihood is modelled to rise with higher patient-to-nurse staffing ratios, lower staff experience, night shift, and high-risk medication classes. These are designed relationships built into the data generator — not measurements of any real health system. Full methodology, bias statement,

and limitations: see `methodology_bias_limitations.md`.

Why This Dataset Is Useful

This dataset gives analysts, students, and QA/software testers a realistic-feeling, richly-structured medication-safety dataset to practise on without any real patient privacy risk. It supports:

- SQL and Python rate-analysis exercises (shift, experience band, staffing ratio)
- Power BI/Tableau dashboard-building practice
- An educational classification exercise (near-miss prediction) with genuine class imbalance
- Portfolio-ready case studies and dashboards — like this one

Restriction / Disclaimer

100% synthetic data. No real patients, clinicians, staff, or facilities are represented. Not clinically validated. For research, education, software testing, and AI/ML training/ evaluation only. Full terms: `LICENSE_AND_ACCEPTABLE_USE.md`.

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